

Static Local Variables & Default Arguments

ELEC1006: ENGINEERING COMPUTING

Static Local Variables

- Local variables only exist while the function is executing. When the function terminates, the contents of local variables are lost.
- `static` local variables retain their contents between function calls.
- `static` local variables are defined and initialized only the first time the function is executed. `0` is the default initialization value.



Example 1: Local Variables

```
1 // This program shows that local variables do not retain
2 // their values between function calls.
3 #include <iostream>
4 using namespace std;
5
6 // Function prototype
7 void showLocal();
8
9 int main()
10 {
11     showLocal();
12     showLocal();
13     return 0;
14 }
15
```

In this program,
each time
showLocal is
called, the
localNum variable
is re-created and
initialized with the
value 5.

```
16 /*******
17 // Definition of function showLocal. *
18 // The initial value of localNum, which is 5, is displayed. *
19 // The value of localNum is then changed to 99 before the *
20 // function returns. *
21 /*******
22
23 void showLocal()
24 {
25     int localNum = 5; // Local variable
26
27     cout << "localNum is " << localNum << endl;
28     localNum = 99;
29 }
```

Program Output

```
localNum is 5
localNum is 5
```

Example 2: Static Local Variables

```
1 // This program uses a static local variable.
2 #include <iostream>
3 using namespace std;
4
5 void showStatic(); // Function prototype
6
7 int main()
8 {
9     // Call the showStatic function five times.
10    for (int count = 0; count < 5; count++)
11        showStatic();
12    return 0;
13 }
14
15 //*****
16 // Definition of function showStatic.
17 // statNum is a static local variable. Its value is displayed
18 // and then incremented just before the function returns.
19 //*****
20
21 void showStatic()
22 {
23     static int statNum;
24
25     cout << "statNum is " << statNum << endl;
26     statNum++;
27 }
```

Program Output

```
statNum is 0
statNum is 1
statNum is 2
statNum is 3
statNum is 4
```

← statNum is automatically initialized to 0. Notice that it retains its value between function calls.

- If you initialize a local static variable, the initialization only **happens once**.
- The lifetime of a static variable is from the moment it is invoked till the end of the program.

```
16  //*****
17  // Definition of function showStatic. *
18  // statNum is a static local variable. Its value is displayed *
19  // and then incremented just before the function returns. *
20  //*****
21
22  void showStatic()
23  {
24      static int statNum = 5;
25
26      cout << "statNum is " << statNum << endl;
27      statNum++;
28  }
```

Program Output

```
statNum is 5
statNum is 6
statNum is 7
statNum is 8
statNum is 9
```

Default Arguments

A Default argument is an argument that is passed automatically to a parameter if the argument is missing on the function call.

- Must be a constant declared in prototype:
`void evenOrOdd(int = 0);`
- Can be declared in header if no prototype
- Multi-parameter functions may have default arguments for some or all of them:

```
int getSum(int, int=0, int=0);
```

Default arguments specified in the prototype

```
1 // This program demonstrates default function arguments.
2 #include <iostream>
3 using namespace std;
4
5 // Function prototype with default arguments
6 void displayStars(int = 10, int = 1);
7
8 int main()
9 {
10     displayStars();           // Use default values for cols and rows.
11     cout << endl;
12     displayStars(5);         // Use default value for rows.
13     cout << endl;
14     displayStars(7, 3);      // Use 7 for cols and 3 for rows.
15     return 0;
16 }
17
18 //*****
19 // Definition of function displayStars.
20 // The default argument for cols is 10 and for rows is 1.
21 // This function displays a square made of asterisks.
22 //*****
23
24 void displayStars(int cols, int rows)
25 {
26     // Nested loop. The outer loop controls the rows
27     // and the inner loop controls the columns.
28     for (int down = 0; down < rows; down++)
29     {
30         for (int across = 0; across < cols; across++)
31             cout << "*";
32         cout << endl;
33     }
34 }
```

Example 3: Default Arguments

Program Output

Default Arguments

- If not all parameters to a function have default values, the **default less** ones are **declared first** in the parameter list:

```
int getSum(int, int=0, int=0); // OK
int getSum(int, int=0, int);   // NO
```

- When an **argument is omitted from a function call**, all arguments **after** it must also be **omitted**:

```
sum = getSum(num1, num2);      // OK
sum = getSum(num1, , num3);    // NO
```


More info

- [1] cplusplus.com: Variables. Data Types.
<https://www.cplusplus.com/doc/oldtutorial/variables/>
- [2] learncpp.com: 6.10 – Static local variables
<https://www.learncpp.com/cpp-tutorial/static-local-variables/>
- [3] learncpp.com: 7.7 – Default arguments
<https://www.learncpp.com/cpp-tutorial/77-default-arguments/>